Adverse Drug Reactions (ADRs) & Pharmacovigilance: Comprehensive Study Guide

1. Introduction & Core Definitions

Adverse Drug Reaction (ADR)

An **Adverse Drug Reaction (ADR)** is defined by the World Health Organization (WHO) as a response to a drug that is noxious and unintended, occurring at doses normally used in humans for prophylaxis, diagnosis, or therapy of disease, or for the modification of physiological function.

- **Adverse Event (AE) vs. Adverse Drug Reaction (ADR):** An Adverse Event is any untoward medical occurrence during treatment with a pharmaceutical product, which does not necessarily have a causal relationship with the treatment. An ADR specifically requires a reasonable possibility of a causal relationship between the drug and the event.
- **Side Effect vs. ADR:** A side effect is a predictable, dose-dependent secondary effect occurring at therapeutic doses, which may be beneficial or harmful. An ADR is strictly untoward, noxious, and harmful to patient health.

Pharmacovigilance (Pv)

Pharmacovigilance is defined as the science and activities relating to the detection, assessment, understanding, and prevention of adverse effects or any other drug-related problems.

- **Scope:** Extends to adverse events caused by prescription medicines, over-the-counter (OTC) drugs, biologicals, herbal remedies, traditional medicines, blood products, medical devices, and vaccines.
- **Core Objectives:**
    - Improve patient care and safety during the use of medicines and medical interventions.
    - Enhance public health and safety regarding the use of medicines.

    - Contribute to the assessment of benefit, harm, effectiveness, and risk of medicines.
    - Promote education, clinical training, and effective communication to the public and healthcare professionals.

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2. Classification of Adverse Drug Reactions

Primary ABCDEF Classification System

- **Type A (Augmented):** Dose-related, predictable from the primary or secondary pharmacology of the drug. Common, low mortality (e.g., Bradycardia from Beta-blockers, Hypoglycemia from Insulin).
- **Type B (Bizarre):** Non-dose related, unpredictable, unrelated to the drug's known pharmacology. Uncommon, high mortality, often immune-mediated or genetic (e.g., Anaphylaxis from Penicillin, SJS/TEN from Anticonvulsants).
- **Type C (Chronic / Continuous):** Dose- and time-related, associated with cumulative long-term therapy (e.g., Adrenal suppression from corticosteroids, Nephrotoxicity from Analgesics).
- **Type D (Delayed):** Time-related, becoming apparent long after drug exposure, often in offspring or later in life (e.g., Teratogenesis, Carcinogenesis).
- **Type E (End of Use):** Occurs soon after withdrawal of a drug (e.g., Rebound hypertension after Clonidine cessation, Opiate withdrawal syndrome).
- **Type F (Failure):** Unexpected failure of therapy (e.g., Resistance to antibiotics, oral contraceptive failure due to enzyme inducers).

Genotoxic & Teratogenic Categorization

- **Genotoxicity:** Damage to cellular genetic material (DNA/chromosomes) causing mutations or chromosomal aberrations, which may lead to malignancy or hereditary defects.
- **Teratogenesis:** Structural or functional defects induced in the developing embryo/fetus when exposed during critical gestational periods.

Category	FDA Pregnancy Risk Description	Example Drugs
Category A	Controlled studies show no risk to fetus in 1st trimester.	Folic Acid, Thyroid Hormone
Category B	Animal studies show no risk, but human studies lacking; or animal studies show risk not confirmed in humans.	Amoxicillin, Metformin
Category C	Animal studies show risk; human studies lacking. Benefits may warrant use.	Lisinopril (1st tri), Fluconazole
Category D	Positive evidence of human fetal risk. Benefits may justify potential risk in life-threatening situations.	Phenytoin, Valproic Acid
Category X	Fetal abnormalities proven in animals/humans. Risks clearly outweigh any potential benefit. Contraindicated.	Thalidomide, Isotretinoin, Warfarin

Allergic & Hypersensitivity Reactions (Gell and Coombs)

Type	Immune Mechanism	Key Mediators / Target	Clinical Manifestations	Example Drugs
Type I (Immediate)	IgE-mediated mast cell degranulation	Histamine, Leukotrienes	Anaphylaxis, Urticaria, Bronchospasm	Penicillins, Cephalosporins
Type II (Cytotoxic)	IgG/IgM binding to cell-bound antigen	Complement activation, Cell lysis	Hemolytic Anemia, Thrombocytopenia	Methyldopa, Quinidine
Type III (Immune Complex)	Antigen-Antibody complex deposition	Complement cascade, Neutrophils	Serum Sickness, Drug-induced Lupus	Sulfonamides, Hydralazine
Type IV (Delayed)	T-cell mediated cellular response	Cytokines, Cytotoxic T-cells	Contact Dermatitis, SJS/TEN, DRESS	Allopurinol, Carbamazepine

Idiosyncratic & Pharmacogenetic Reactions

Unpredictable, non-immunological ADRs occurring in genetically susceptible individuals:

- **G6PD Deficiency:** Exposure to oxidative drugs (e.g., Primaquine, Sulfonamides, Rasburicase) causes acute hemolytic anemia.
- **Malignant Hyperthermia:** Genetic mutations in the $RyR1$ receptor cause hypermetabolic crises when exposed to volatile anesthetics (e.g., Halothane) or Succinylcholine.
- **Slow Acetylators ($NAT2$ polymorphism):** Increased risk of Hydralazine-induced lupus or Isoniazid-induced peripheral neuropathy.

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3. Causality Assessment Tools

Bradford Hill Criteria (Epidemiological Causality)

Used to evaluate whether an observed association between an exposure and an outcome is causal:

1. **Strength of Association:** Larger effect size increases likelihood of causation.
2. **Consistency:** Repeated observations in different populations and settings.
3. **Specificity:** Specific exposure causing a specific outcome.
4. **Temporality:** Cause must precede the effect (essential criterion).
5. **Biological Gradient:** Dose-response relationship.
6. **Plausibility:** Biologically plausible mechanism.
7. **Coherence:** Alignment with known human biology and disease pathology.
8. **Experiment:** Experimental evidence (e.g., drug withdrawal/rechallenge).
9. **Analogy:** Similarity with known effects of similar drugs.

WHO-UMC Causality Assessment Scale

- **Certain:** Clinical event, including lab test abnormality, with plausible time relationship to drug intake. Cannot be explained by disease or other drugs. Response to withdrawal (dechallenge) plausible. Event definitive pharmacologically/phenomenologically; rechallenge positive.

- **Probable / Likely:** Clinical event with reasonable time sequence. Unlikely to be attributed to disease or other drugs. Dechallenge plausible. Rechallenge not required.
- **Possible:** Clinical event with reasonable time sequence. Could also be explained by disease or other drugs. Information on drug withdrawal may be lacking or unclear.
- **Unlikely:** Clinical event with temporal relationship that makes a causal relationship improbable. Other drugs, chemicals, or underlying disease provide plausible explanations.
- **Conditional / Unclassified:** Event or lab result reported; more data needed for proper assessment.
- **Unassessable / Unclassifiable:** Report suggesting an adverse reaction, but cannot be judged due to insufficient or contradictory information.

Naranjo Causality Scale (Quantitative Algorithm)

Evaluation Question	Yes	No	Do Not Know
1. Are there previous conclusive reports on this reaction?	+1	0	0
2. Did the adverse event appear after the suspected drug was administered?	+2	-1	0
3. Did the adverse reaction improve when the drug was discontinued or a specific antagonist was administered?	+1	0	0
4. Did the adverse reaction reappear when the drug was re-administered?	+2	-1	0
5. Are there alternative causes (other than the drug) that could on their own have caused this reaction?	-1	+2	0

Evaluation Question	Yes	No	Do Not Know
6. Did the reaction reappear when a placebo was given?	-1	+1	0
7. Was the drug detected in the blood (or other fluids) in concentrations known to be toxic?	+1	0	0
8. Was the reaction more severe when the dose was increased, or less severe when the dose was decreased?	+1	0	0
9. Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	+1	0	0
10. Was the adverse event confirmed by any objective evidence?	+1	0	0

Naranjo Total Score Interpretation

- **≥ 9:** Definite ADR
- **$5 - 8$:** Probable ADR
- **$1 - 4$:** Possible ADR
- **≤ 0:** Unlikely ADR

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4. ADR Reporting & Pharmacovigilance Framework

Pharmacovigilance Programme of India (PvPI)

- **Initiation:** Launched in July 2010 by the Ministry of Health & Family Welfare, Government of India.

- **National Coordinating Centre (NCC):** Indian Pharmacopoeia Commission (IPC), Ghaziabad.
- **Operational Framework:** Works under the administrative control of the Central Drugs Standard Control Organization (CDSCO). Coordinates hundreds of Adverse Drug Reaction Monitoring Centres (AMCs) across medical colleges and hospitals nationwide.

Reporting Mechanisms

1. **CDSCO Red Suspected ADR Reporting Form:** Official paper-based form used by healthcare professionals (doctors, pharmacists, nurses) to log adverse events.
2. **ADR PvPI Mobile App:** Digital platform enabling rapid, point-of-care reporting by healthcare providers and consumers.
3. **Toll-Free Helpline:** Direct telephonic reporting via national helpline numbers.

Practical Workflow: Clinical Scenario Analysis

Clinical Presentation & Patient Timeline

A 52-year-old male with a history of hypertension presented to the outpatient clinic.

- **Day 1:** Initiated on **Enalapril 10 mg PO daily** for blood pressure management.
- **Day 10:** Patient developed a persistent, dry, non-productive cough. No fever, dyspnea, or signs of upper respiratory infection.
- **Day 14:** Cough worsened, interfering with sleep. Physical examination and chest X-ray were unremarkable. Enalapril was discontinued (**Dechallenge**), and the patient was switched to Amlodipine 5 mg PO daily.
- **Day 18:** Dry cough completely resolved within 4 days of Enalapril cessation.

Causality Assessment & Form Completion Data

[PATIENT INFORMATION]

Patient Initials: J.D. Age: 52 Gender: Male Weight: 72 kg

[SUSPECTED ADVERSE REACTION]

Description: Persistent dry, non-productive cough without systemic infection.

Date of Reaction Onset: Day 10 post-initiation

Outcome: Recovered fully after drug cessation.

[SUSPECTED DRUG(S)]

1. Drug Name: Enalapril

   Dose: 10 mg PO QD

   Indication: Essential Hypertension

   Therapy Dates: [Start Date] to [End Date]

   Action Taken: Drug Discontinued (Positive Dechallenge)

[CAUSALITY ASSESSMENT (Naranjo Algorithm Analysis)]

- Conclusive reports on reaction? Yes (+1) [Known ACE inhibitor effect via Bradykinin]

- Reaction after drug intake? Yes (+2)

- Reaction improved upon discontinuation? Yes (+1)

- Alternative causes present? No (+2)

- Objective evidence? Clinical resolution documented (+1)

TOTAL NARANJO SCORE: 7 → PROBABLE ADR

Regulatory Submission Steps

1. **Identification:** The treating physician or clinical pharmacist identifies the clinical event as a potential ADR during routine assessment.
2. **Data Collection:** Gather patient demographic details, complete drug history, timeline, clinical intervention, and outcome.
3. **Documentation:** Complete the CDSCO Suspected ADR Form or input the details into the ADR PvPI mobile application.

4. **Transmission:** Submit the documented report to the local Adverse Drug Reaction Monitoring Centre (AMC) or upload directly to the NCC-IPC database.
5. **Signal Detection:** Data is integrated into VigiBase (WHO Uppsala Monitoring Centre) for global signal detection and continuous safety evaluation.